

DERA ISMAIL KHAN, Pakistan

WMO: 416240

Lat: 31.909N

Lon: 70.897E

Elev: 594

StdP: 14.38

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.4	42.5	23.6	18.3	60.3	29.9	24.5	54.5	11.4	58.2	8.4	57.9	0.4	N/A	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.3	109.3	78.6	106.7	78.9	104.2	79.9	87.1	99.9	85.9	98.7	85.0	97.5	0.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.2	183.1	96.3	83.0	176.0	94.9	81.9	169.9	93.7	53.1	100.2	51.5	98.9	50.4	98.9	96.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.8	6.7	5.0		DB	37.0	114.0	2.3	2.2	35.4	115.6	34.0	116.8	32.7	118.0	31.0	119.6
				WB	34.8	89.6	2.4	2.4	33.1	91.3	31.7	92.7	30.4	94.0	28.6	95.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	54.7	60.7	69.7	81.4	90.3	92.9	91.1	89.7	87.2	79.4	67.2	57.8
	DBStd	14.32	4.23	4.63	6.44	5.93	4.75	4.62	4.00	3.88	3.56	4.79	4.02	4.44
	HDD50	17	11	0	0	0	0	0	0	0	0	0	0	6
	HDD65	725	321	133	27	0	0	0	0	0	0	0	21	223
	CDD50	9846	155	301	611	942	1250	1288	1273	1231	1116	913	517	249
	CDD65	5079	0	14	173	492	785	838	808	766	666	448	88	1
	CDH74	70922	8	137	1507	5807	11933	13368	12457	11276	9067	4611	695	56
	CDH80	43175	0	14	530	3220	7817	9136	8054	6920	5121	2239	123	1
Wind	WSAvg	1.0	0.8	1.2	1.1	1.2	1.4	1.5	1.4	1.0	0.8	0.6	0.6	0.7
Precipitation	PrecAvg	14.0	0.4	1.0	1.4	1.0	0.4	1.1	3.0	3.3	1.6	0.4	0.1	0.2
	PrecMax	29.3	2.4	4.7	4.1	2.9	1.5	3.9	8.8	14.5	4.9	3.0	1.3	1.1
	PrecMin	6.1	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0
	PrecStd	5.4	0.6	1.1	1.0	0.9	0.4	1.0	2.4	3.3	1.4	0.7	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.0	82.0	94.2	106.2	112.7	113.0	108.4	104.5	102.5	98.6	86.4	78.7
		MCWB	58.6	64.1	72.2	78.3	75.6	80.4	83.7	86.6	84.4	78.7	68.8	62.5
	2%	DB	71.6	77.9	89.7	101.7	109.2	109.9	104.7	102.4	100.4	96.1	83.4	75.3
		MCWB	57.3	62.5	71.5	74.4	76.0	79.4	85.0	85.4	82.2	77.0	67.1	60.1
	5%	DB	68.6	75.1	85.9	98.3	106.4	107.5	102.4	100.4	98.5	93.7	80.8	72.6
		MCWB	55.5	60.4	69.7	72.6	75.9	79.7	84.8	84.7	80.7	75.4	65.8	58.9
	10%	DB	65.8	71.9	82.4	94.7	103.6	104.8	100.2	98.4	96.6	90.8	78.3	69.5
		MCWB	54.4	59.3	67.3	71.7	75.4	79.7	84.6	84.3	80.3	73.5	64.7	57.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	67.3	75.9	81.0	82.7	87.0	88.7	90.0	85.9	81.4	72.1	65.7
		MCDB	71.4	77.9	90.2	99.2	103.9	106.2	101.5	100.8	99.2	95.5	82.3	74.8
	2%	WB	59.0	64.9	73.5	78.2	80.4	85.2	87.2	87.1	84.6	79.4	69.7	62.8
		MCDB	69.2	74.9	86.3	98.0	102.5	103.0	99.6	99.0	97.1	93.1	80.3	72.6
	5%	WB	57.3	63.0	71.1	75.8	78.8	83.8	86.1	85.8	83.2	77.5	67.8	60.6
		MCDB	66.5	72.4	83.3	93.8	101.0	100.8	98.4	97.5	95.0	90.5	77.8	70.1
	10%	WB	55.7	60.9	68.9	73.4	77.4	82.3	85.1	84.9	81.7	75.7	66.0	58.5
		MCDB	63.9	69.6	80.2	89.9	98.6	98.1	97.4	95.9	93.1	87.5	75.7	67.4
Mean Daily Temperature Range	5% DB	MDBR	21.1	20.1	19.5	23.0	24.2	21.3	16.5	15.8	17.8	21.7	22.5	22.6
		MCDBR	25.9	25.5	24.1	27.7	27.8	25.1	19.4	18.3	19.8	22.9	25.0	27.4
		MCWBR	15.1	13.3	10.9	10.7	9.3	8.8	6.6	6.8	7.1	9.0	13.0	16.7
	5% WB	MCDBR	23.5	21.7	23.0	26.5	25.7	20.8	17.4	17.0	18.2	21.5	22.2	25.5
		MCWBR	14.9	12.3	11.2	11.7	10.1	8.5	7.5	7.6	7.6	9.4	13.1	16.7
Clear-Sky Solar Irradiance	taub	0.451	0.494	0.516	0.563	0.607	0.702	0.796	0.743	0.632	0.610	0.566	0.476	
	taud	1.929	1.875	1.853	1.787	1.705	1.565	1.453	1.538	1.688	1.650	1.693	1.866	
	Ebn at Noon	232	236	243	238	228	207	187	194	210	200	193	214	
	Edh at Noon	49	57	62	69	76	87	97	88	73	71	62	50	
All-Sky Solar Radiation	RadAvg	990	1240	1629	2005	2229	2170	1888	1794	1710	1459	1112	972	
	RadStd	72	119	115	91	71	93	80	100	86	58	64	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page